

Figure 3: Kotlin in Android

If you wish to install the Kotlin compiler in your system, try the following command from a terminal (for Ubuntu users):

```
$ sudo snap install --classic kotlin
```

Detailed instructions for command line installation for other OSs are available at <https://kotlinlang.org/docs/tutorials/command-line.html>.

Kotlin in Android

The official Kotlin documentation lists the major features that Kotlin offers for Android development.

- **Compatibility:** The compatibility with JDK 6 ensures that Kotlin based applications can run on older Android devices.
- **Kotlin is 100 per cent interoperable with Java:** This means that all the existing Android libraries can be used in Kotlin applications.
- **A minimal footprint:** This important feature is available because Kotlin has a very compact run-time library.

If you are a Java developer, you can get started with Kotlin instantly. There is an automated Java-to-Kotlin converter available.

Kotlin has been used by major applications such as Pinterest, Basecamp's Android app, etc. For sample Android apps built using Kotlin, you can go to <https://developer.android.com/samples/index.html?language=kotlin>. Google has provided some interesting applications built using Kotlin at this link.

A detailed tutorial on getting started with Kotlin for Android app development is available at <https://kotlinlang.org/docs/tutorials/kotlin-android.html>.

Kotlin in browsers

As stated earlier, Kotlin enables the developer to transpile Kotlin code to JavaScript. The Kotlin-to-JavaScript compiler ensures that the code generated is optimal in size, and readability is also ensured.

Some of the prime reasons for targeting JavaScript transpiling are given below.

- **DOM interaction:** With the features provided by Kotlin, DOM can be handled. It is possible to update existing elements and create new elements too.
- **The graphics interaction can be managed through**

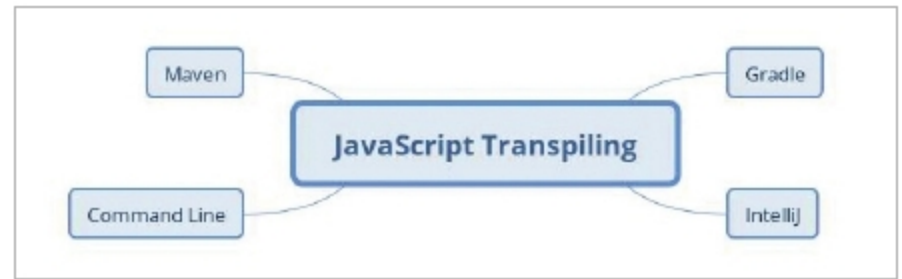


Figure 4: JavaScript transpiling

WebGL: This can be used to build graphical components in the Web pages.

Kotlin can be adopted for browser-side use along with other JavaScript libraries or frameworks such as jQuery or React.

The conversion from Kotlin to JavaScript can be done in multiple ways. The official documentation has listed the following methods:

- Gradle approach
- IntelliJ IDEA approach
- Maven approach
- Command line approach.

In the 'Kotlin – Try online' tutorial, the 'Hello World' example produces the following JavaScript code:

```
kotlin.kotlin.io.output.flush();
if (typeof kotlin === 'undefined')
{
  throw new Error("Error loading module 'moduleId'. Its
  dependency 'kotlin' was not found. Please, check whether
  'kotlin' is loaded prior to 'moduleId'.");
}
var moduleId = function (_, Kotlin)
{ 'use strict';
  var println = Kotlin.kotlin.io.println_s8jyv4$;
  function main(args)
  { println('Hello, world! Sample for OSFY'); } _._main_kand9s$
  = main; main([]); Kotlin.defineModule('moduleId', _); return
  _; }(typeof moduleId === 'undefined' ? {} : moduleId, kotlin);
kotlin.kotlin.io.output.buffer;
```

The detailed instructions on how to handle the aforementioned approaches are given at <https://kotlinlang.org/docs/tutorials/javascript/kotlin-to-javascript/kotlin-to-javascript.html>.

Native Kotlin

Kotlin has a feature for compiling to native binaries, and can execute this feature without any virtual machine. The target platforms that native Kotlin supports are listed below:

- Windows (x86_64)
- Linux
- iOS
- Android
- Web Assembly